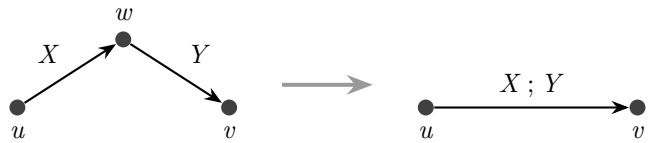
An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

Yuya Uezato Yoshiki Nakamura

CyberAgent, Inc. Chiba University

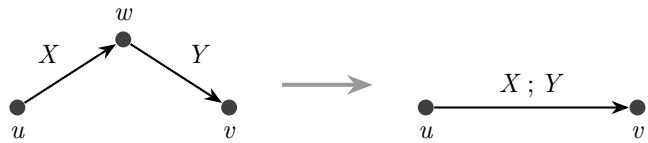
STOC 2027 submission (author video for the PC)

The calculus of relations (CoR)



$$X\,;\,Y \;\coloneqq\; \{(u,v) : (u,w) \in X \text{ and } (w,v) \in Y \text{ for some } w\}$$

The calculus of relations (CoR)



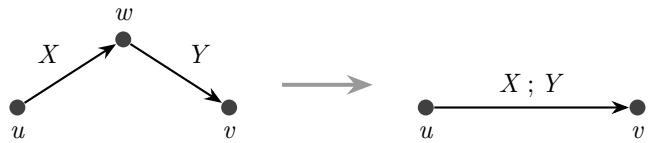
$$X\,;\,Y \;\coloneqq\; \{(u,v) : (u,w) \in X \text{ and } (w,v) \in Y \text{ for some } w\}$$

Terms over binary relation symbols, built from

$$\cup, \quad \cap, \quad \bar{\cdot} \text{ (complement)}, \quad \breve{\cdot} \text{ (converse)}, \quad \mathbf{0},\ \mathbf{1},\ \mathbf{1}', \quad \text{and} \quad ;$$

Every intermediate result is a **binary** relation.

The calculus of relations (CoR)



$$X \,;\, Y \;\coloneqq\; \{(u,v) : (u,w) \in X \text{ and } (w,v) \in Y \text{ for some } w\}$$

Terms over binary relation symbols, built from

$$\cup, \quad \cap, \quad \overline{\cdot} \text{ (complement)}, \quad \breve{} \text{ (converse)}, \quad \mathbf{0},\ \mathbf{1},\ \mathbf{1}', \quad \text{and} \quad ;$$

Every intermediate result is a **binary** relation.

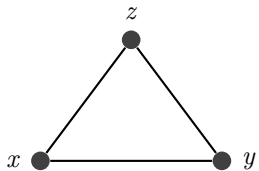
As a matrix program

; = Boolean matrix product
$\breve{}$ = transpose, others entrywise

As a query plan

; = join followed by projection
(Codd 1970), (Brijder et al. 2019)

Three-variable logic and CoR



FO^3: first-order logic with the variables x, y, z only

$\equiv$

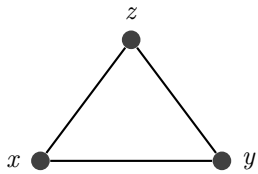


CoR: binary relations only

Theorem (Tarski; (Tarski and Givant 1987)**)**

FO^3 *formulas* $\varphi(x, y)$ *and* CoR *terms define the same binary queries.*

Three-variable logic and CoR



FO^3: first-order logic with the variables x, y, z only

$\equiv$



CoR: binary relations only

Theorem (Tarski; (Tarski and Givant 1987)**)**

FO^3 *formulas* $\varphi(x, y)$ *and* CoR *terms define the same binary queries.*

$\mathrm{CoR} \to \mathrm{FO}^3$	size $O(\lvert T\rvert)$	composition $\mapsto \exists z$
$\mathrm{FO}^3 \to \mathrm{CoR}$	size $2^{O(\lvert\varphi\rvert)}$	classical translation

Why translate: fast evaluation (Nakamura and Uezato 2026)

$\mathrm{FO}^k,\ k \geq 4$	no $O(m^{k-\varepsilon})$ algorithm (under Sparse MAX-3-SAT)
FO^3	$\xrightarrow{\text{translate}}\ \mathrm{CoR}\ \xrightarrow{\text{matrix mult.}}\ 2^{O(\lvert\varphi\rvert)} \cdot n^{\omega}$
FO^2	$O(\lvert\varphi\rvert \cdot m)$

n: domain size, m: number of tuples, naive evaluation of FO^k: $O(|\varphi| \cdot n^k)$

Why translate: fast evaluation (Nakamura and Uezato 2026)

$\mathrm{FO}^k,\ k \geq 4$	no $O(m^{k-\varepsilon})$ algorithm (under Sparse MAX-3-SAT)
FO^3	$\xrightarrow{\text{translate}}\ \mathrm{CoR}\ \xrightarrow{\text{matrix mult.}}\ 2^{O(\lvert\varphi\rvert)} \cdot n^\omega$
FO^2	$O(\lvert\varphi\rvert \cdot m)$

n: domain size, m: number of tuples, naive evaluation of FO^k: $O(|\varphi| \cdot n^k)$

A CoR term T is evaluated in $O(|T| \cdot n^\omega)$:
one matrix product per composition.

Why translate: fast evaluation (Nakamura and Uezato 2026)

$\mathrm{FO}^k,\ k \geq 4$	no $O(m^{k-\varepsilon})$ algorithm (under Sparse MAX-3-SAT)
FO^3	$\xrightarrow{\text{translate}}\ \mathrm{CoR}\ \xrightarrow{\text{matrix mult.}}\ 2^{O(\lvert\varphi\rvert)} \cdot n^\omega$
FO^2	$O(\lvert\varphi\rvert \cdot m)$

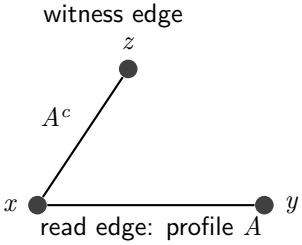
n: domain size, m: number of tuples, naive evaluation of FO^k: $O(|\varphi| \cdot n^k)$

A CoR term T is evaluated in $O(|T| \cdot n^\omega)$:
one matrix product per composition.

The factor $2^{O(|\varphi|)}$ comes only from the translation $\mathrm{FO}^3 \to \mathrm{CoR}$.
Is it necessary?

Where the exponential comes from

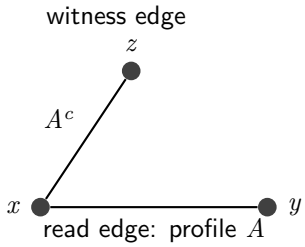
$$G_k(x,y) := \exists z \bigwedge_{i=1}^{k} \big[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z))\big] \qquad \text{size } O(k)$$



profile of an edge $= \{i : P_i \text{ holds}\}$
(on inputs where exactly one of P_i, Q_i holds)

Where the exponential comes from

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \big[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \big] \qquad \text{size } O(k)$$



profile of an edge $= \{i : P_i \text{ holds}\}$
(on inputs where exactly one of P_i, Q_i holds)

The witness edge must have the **complementary** profile of the read edge.

Composition accesses the witness edge separately,
so the classical translation **fixes** the read profile:

$$G_k \;\equiv\; \bigcup_{A \subseteq [k]} \big[\, H_A \cap (H_{A^c}\,;\mathbf{1}) \,\big]$$

$H_A \coloneqq \bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i$ $\qquad$ 2^k **compositions**

Is the blow-up necessary?

target	complement	exponential lower bound
positive CoR	none	known (Nakamura 2022)
$\Sigma_1\mathrm{CoR}$	on inputs only	known (Nakamura 2022)
full CoR	anywhere	open

Full CoR = Boolean matrix programs with arbitrary Boolean operations, transpose, and products.

Is the blow-up necessary?

target	complement	exponential lower bound
positive CoR	none	known (Nakamura 2022)
$\Sigma_1\mathrm{CoR}$	on inputs only	known (Nakamura 2022)
full CoR	anywhere	**this work**

Full CoR = Boolean matrix programs with arbitrary Boolean operations, transpose, and products.

Main result

Theorem

For every $k \geq 1$, every relational circuit C with $C \equiv_{\mathrm{fin}} G_k$ has

$$\#\textit{composition gates of } C \;\geq\; 2^{k-1}.$$

- circuit: shared subterms are counted once
- $\cup, \cap$, complement, converse, $\mathbf{0}, \mathbf{1}, \mathbf{1}'$: unrestricted and not counted
- $\equiv_{\mathrm{fin}}$: equivalence on all **finite** structures

Main result

Theorem

For every $k \geq 1$, every relational circuit C with $C \equiv_{\mathrm{fin}} G_k$ has

$$\#\textit{composition gates of } C \;\geq\; 2^{k-1}.$$

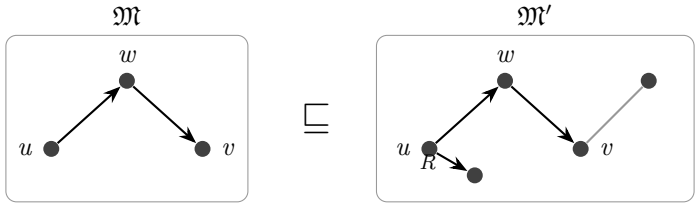
- circuit: shared subterms are counted once
- $\cup, \cap$, complement, converse, $\mathbf{0}, \mathbf{1}, \mathbf{1}'$: unrestricted and not counted
- $\equiv_{\mathrm{fin}}$: equivalence on all **finite** structures

Consequences

- 2^k compositions suffice: the classical $2^{O(|\varphi|)}$ translation is optimal.
- Evaluating FO^3 through CoR needs $2^{\Omega(|\varphi|)}$ Boolean matrix products.

This bounds whole-matrix operations, not the running time of arbitrary algorithms (G_k is directly evaluable in $O(kn^3)$).

The known method needs monotonicity

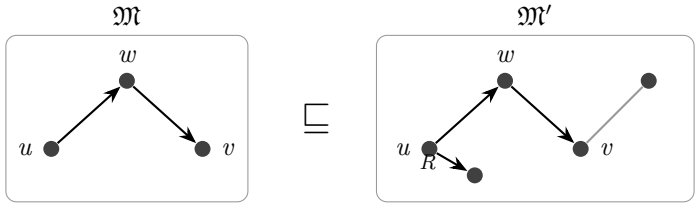


Fact (for $\Sigma_1\mathrm{CoR}$)

If $\mathfrak{M}$ is an induced substructure of $\mathfrak{M}'$, then $T^{\mathfrak{M}} \subseteq T^{\mathfrak{M}'}$.

A witness in $\mathfrak{M}$ remains a witness in $\mathfrak{M}'$. (Nakamura 2022) combines this with rectangularity of ; (a fooling-set argument).

The known method needs monotonicity



Fact (for $\Sigma_1\mathrm{CoR}$)

If $\mathfrak{M}$ is an induced substructure of $\mathfrak{M}'$, then $T^{\mathfrak{M}} \subseteq T^{\mathfrak{M}'}$.

A witness in $\mathfrak{M}$ remains a witness in $\mathfrak{M}'$. (Nakamura 2022) combines this with rectangularity of ; (a fooling-set argument).

Complement reverses inclusion

$(u, u) \in \overline{R\,;\mathbf{1}}$ iff u has no outgoing R-edge: true in $\mathfrak{M}$, false in $\mathfrak{M}'$.

Symmetric structures are infinite

Infinite homogeneous colored graph

Every CoR-definable relation is invariant under automorphisms, hence a union of color classes (and possibly $\mathbf{1}'$).

Complement is harmless.

Symmetric structures are infinite

Infinite homogeneous colored graph

Every CoR-definable relation is invariant under automorphisms, hence a union of color classes (and possibly $\mathbf{1}'$).

Complement is harmless.

Correctness on finite structures

An infinite counter-model does not exclude a short term that is correct on all **finite** structures.

Finite homogeneous structures are rare.

Symmetric structures are infinite

Infinite homogeneous colored graph

Every CoR-definable relation is invariant under automorphisms, hence a union of color classes (and possibly $\mathbf{1}'$).

Complement is harmless.

Correctness on finite structures

An infinite counter-model does not exclude a short term that is correct on all **finite** structures.

Finite homogeneous structures are rare.

We need a comparison that is preserved by complement,
carried out on a **finite** structure.

Proof outline

1. A finite structure $\mathfrak{M}_0$ and 2^k structures $\mathfrak{M}_A$ $(A \subseteq [k])$ on the same domain, with
$$G_k^{\mathfrak{M}_0}(h, v_A) = 1, \qquad G_k^{\mathfrak{M}_A}(h, v_A) = 0,$$
where $\mathfrak{M}_A$ differs from $\mathfrak{M}_0$ only on edges incident to h, and not on (h, v_A).

Proof outline

1. A finite structure $\mathfrak{M}_0$ and 2^k structures $\mathfrak{M}_A$ $(A \subseteq [k])$ on the same domain, with
$$G_k^{\mathfrak{M}_0}(h, v_A) = 1, \qquad G_k^{\mathfrak{M}_A}(h, v_A) = 0,$$
where $\mathfrak{M}_A$ differs from $\mathfrak{M}_0$ only on edges incident to h, and not on (h, v_A).

2. For a circuit C, compare each gate on $\mathfrak{M}_A$ with a value computed from $\mathfrak{M}_0$, by **equality**.

Proof outline

1. A finite structure $\mathfrak{M}_0$ and 2^k structures $\mathfrak{M}_A$ $(A \subseteq [k])$ on the same domain, with
$$G_k^{\mathfrak{M}_0}(h, v_A) = 1, \qquad G_k^{\mathfrak{M}_A}(h, v_A) = 0,$$
where $\mathfrak{M}_A$ differs from $\mathfrak{M}_0$ only on edges incident to h, and not on (h, v_A).

2. For a circuit C, compare each gate on $\mathfrak{M}_A$ with a value computed from $\mathfrak{M}_0$, by **equality**.

3. Each composition gate breaks this equality for at most **two** sets A.

Proof outline

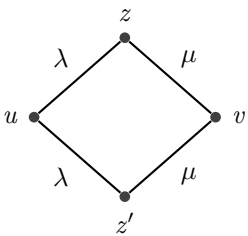
1. A finite structure $\mathfrak{M}_0$ and 2^k structures $\mathfrak{M}_A$ $(A \subseteq [k])$ on the same domain, with
$$G_k^{\mathfrak{M}_0}(h, v_A) = 1, \qquad G_k^{\mathfrak{M}_A}(h, v_A) = 0,$$
where $\mathfrak{M}_A$ differs from $\mathfrak{M}_0$ only on edges incident to h, and not on (h, v_A).

2. For a circuit C, compare each gate on $\mathfrak{M}_A$ with a value computed from $\mathfrak{M}_0$, by **equality**.

3. Each composition gate breaks this equality for at most **two** sets A.

$$2^k \;\leq\; 2 \cdot \#\text{composition gates of } C.$$

A finite colored complete graph

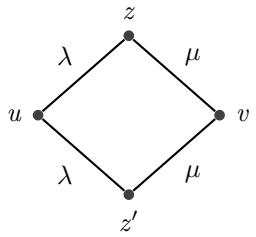


Lemma

For $n \geq O(N^2 \log N)$ there is a coloring χ of the edges of K_n with N colors such that for all $u \neq v$ and all colors λ, μ, at least two $z \notin \{u, v\}$ satisfy $\chi(u, z) = \lambda$ and $\chi(z, v) = \mu$.

Random coloring and a union bound; cf. (Jipsen et al. 1995).

A finite colored complete graph



Lemma
For $n \geq O(N^2 \log N)$ there is a coloring χ of the edges of K_n with N colors such that for all $u \neq v$ and all colors λ, μ, at least two $z \notin \{u, v\}$ satisfy $\chi(u, z) = \lambda$ and $\chi(z, v) = \mu$.

Random coloring and a union bound; cf. (Jipsen et al. 1995).

A **color table** $p = (p_{\mathbf{1}'}, (p_\lambda)_\lambda)$ denotes the relation that is $p_{\mathbf{1}'}$ on the diagonal and $p_{\chi(u,v)}$ at (u, v), $u \neq v$.

Lemma
Color tables are closed under all CoR *operations. For λ,*

$$(p\,;\,q)_\lambda \;=\; (p_{\mathbf{1}'} \wedge q_\lambda) \vee (p_\lambda \wedge q_{\mathbf{1}'}) \vee \Big(\bigvee\nolimits_\mu p_\mu\Big) \wedge \Big(\bigvee\nolimits_\nu q_\nu\Big).$$

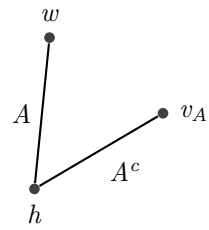
The structures $\mathfrak{M}_0$ and $\mathfrak{M}_A$

Colors $= 2^{[k]}$. On an edge of color μ:

$$P_i \iff i \in \mu, \qquad Q_i \iff i \notin \mu.$$

On loops, all P_i, Q_i are false.

At (x, y) of color A^c, a witness z for G_k must have $\chi(x, z) = A$.



$\mathfrak{M}_0$

The structures $\mathfrak{M}_0$ and $\mathfrak{M}_A$

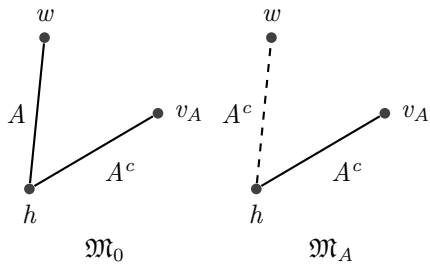
Colors $= 2^{[k]}$. On an edge of color μ:

$$P_i \iff i \in \mu, \qquad Q_i \iff i \notin \mu.$$

On loops, all P_i, Q_i are false.

At (x, y) of color A^c, a witness z for G_k must have $\chi(x, z) = A$.

$\mathfrak{M}_A$: fix a vertex h and recolor every edge $\{h, w\}$ of color A to A^c.



The structures $\mathfrak{M}_0$ and $\mathfrak{M}_A$

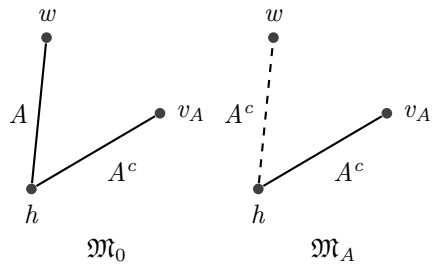
Colors $= 2^{[k]}$. On an edge of color μ:

$$P_i \iff i \in \mu, \qquad Q_i \iff i \notin \mu.$$

On loops, all P_i, Q_i are false.

At (x, y) of color A^c, a witness z for G_k must have $\chi(x, z) = A$.

$\mathfrak{M}_A$: fix a vertex h and recolor every edge $\{h, w\}$ of color A to A^c.



Lemma
$G_k^{\mathfrak{M}_0}(h, v_A) = 1$ *and* $G_k^{\mathfrak{M}_A}(h, v_A) = 0$; $\chi(h, v_A) = A^c$ *in both.*

Comparing the two computations

For a color table p, let $T_A p$ be the relation on $\mathfrak{M}_A$ given by the same table:

$$(T_A p)(u, v) \coloneqq p_{\chi_A(u,v)} \quad (u \neq v), \qquad (T_A p)(u, u) \coloneqq p_{\mathbf{1}'},$$

where χ_A is the coloring of $\mathfrak{M}_A$.

Comparing the two computations

For a color table p, let $T_A p$ be the relation on $\mathfrak{M}_A$ given by the same table:

$$(T_A p)(u, v) \coloneqq p_{\chi_A(u,v)} \quad (u \neq v), \qquad (T_A p)(u, u) \coloneqq p_{\mathbf{1}'},$$

where χ_A is the coloring of $\mathfrak{M}_A$.

- Each input relation of $\mathfrak{M}_A$ is T_A of the same input relation of $\mathfrak{M}_0$.
- T_A commutes with $\cup$, $\cap$, complement, converse, $\mathbf{0}, \mathbf{1}, \mathbf{1}'$.
- T_A need not commute with composition.

Comparing the two computations

For a color table p, let $T_A p$ be the relation on $\mathfrak{M}_A$ given by the same table:

$$(T_A p)(u, v) \coloneqq p_{\chi_A(u,v)} \quad (u \neq v), \qquad (T_A p)(u, u) \coloneqq p_{\mathbf{1}'},$$

where χ_A is the coloring of $\mathfrak{M}_A$.

- Each input relation of $\mathfrak{M}_A$ is T_A of the same input relation of $\mathfrak{M}_0$.
- T_A commutes with $\cup$, $\cap$, complement, converse, $\mathbf{0}, \mathbf{1}, \mathbf{1}'$.
- T_A need not commute with composition.

Lemma

Let C be a circuit. If $(T_A X)\,;(T_A Y) = T_A(X\,;Y)$ for the values X, Y on $\mathfrak{M}_0$ of the operands of every composition gate, then $C(\mathfrak{M}_A) = T_A\, C(\mathfrak{M}_0)$. In particular $C(\mathfrak{M}_A)(h, v_A) = C(\mathfrak{M}_0)(h, v_A)$.

Key lemma

Lemma

For all color tables p, q*:* $\left|\{A : (T_A p)\,;(T_A q) \neq T_A(p\,;q)\}\right| \leq 2.$

Key lemma

Lemma

For all color tables p, q: $\left|\{A : (T_A p)\,;(T_A q) \neq T_A(p\,;q)\}\right| \leq 2.$

Let $J = \text{colors} \setminus \{A\}$ (the colors at h in $\mathfrak{M}_A$) and $\exists_S r \coloneqq \bigvee_{\lambda \in S} r_\lambda$. The two sides can differ only at

pair	$(T_A p)\,;(T_A q)$	$T_A(p\,;q)$
$(h, v),\ v \neq h$	$\exists_J\, p \wedge \exists_{\text{all}}\, q$	$\exists_{\text{all}}\, p \wedge \exists_{\text{all}}\, q$
$(u, h),\ u \neq h$	$\exists_{\text{all}}\, p \wedge \exists_J\, q$	$\exists_{\text{all}}\, p \wedge \exists_{\text{all}}\, q$
(h, h)	$\exists_J\,(p \wedge q)$	$\exists_{\text{all}}\,(p \wedge q)$

Other terms coincide; away from h, the second witness avoids h.

Key lemma

Lemma

For all color tables p, q: $\big|\{A : (T_A p)\,;(T_A q) \neq T_A(p\,;q)\}\big| \leq 2.$

Let $J = \text{colors} \setminus \{A\}$ (the colors at h in $\mathfrak{M}_A$) and $\exists_S r \coloneqq \bigvee_{\lambda \in S} r_\lambda$. The two sides can differ only at

pair	$(T_A p)\,;(T_A q)$	$T_A(p\,;q)$
$(h, v),\ v \neq h$	$\exists_J p \wedge \exists_{\text{all}} q$	$\exists_{\text{all}} p \wedge \exists_{\text{all}} q$
$(u, h),\ u \neq h$	$\exists_{\text{all}} p \wedge \exists_J q$	$\exists_{\text{all}} p \wedge \exists_{\text{all}} q$
(h, h)	$\exists_J (p \wedge q)$	$\exists_{\text{all}} (p \wedge q)$

Other terms coincide; away from h, the second witness avoids h.

Removing one color A changes $\exists_S$ only if A is the unique λ with $r_\lambda = 1$. If some λ has $p_\lambda = q_\lambda = 1$, only $A = \lambda$ matters; otherwise only the unique p-color and the unique q-color.

Counting

For each composition gate g with operand values X_g, Y_g on $\mathfrak{M}_0$, let

$$D_g \coloneqq \{A : (T_A X_g)\,;\,(T_A Y_g) \neq T_A(X_g\,;\,Y_g)\}, \qquad |D_g| \leq 2.$$

Counting

For each composition gate g with operand values X_g, Y_g on $\mathfrak{M}_0$, let

$$D_g \coloneqq \{A : (T_A X_g)\,;\,(T_A Y_g) \neq T_A(X_g\,;\,Y_g)\}, \qquad |D_g| \leq 2.$$

Theorem

If C is correct on $\mathfrak{M}_0$ and on every $\mathfrak{M}_A$, then every $A \subseteq [k]$ lies in some D_g. Hence

$$2^k \;\leq\; \Big|\bigcup_g D_g\Big| \;\leq\; 2 \cdot \#\textit{composition gates of } C.$$

Counting

For each composition gate g with operand values X_g, Y_g on $\mathfrak{M}_0$, let

$$D_g \coloneqq \{A : (T_A X_g)\,;\,(T_A Y_g) \neq T_A(X_g\,;\,Y_g)\}, \qquad |D_g| \leq 2.$$

Theorem

If C is correct on $\mathfrak{M}_0$ and on every $\mathfrak{M}_A$, then every $A \subseteq [k]$ lies in some D_g. Hence

$$2^k \;\leq\; \Big|\bigcup_g D_g\Big| \;\leq\; 2 \cdot \#\textit{composition gates of } C.$$

- Complement: the invariant is an equality, preserved by every Boolean operation.
- Sharing: D_g is defined per gate, so a shared gate is counted once.

Summary

Question CoR evaluates FO^3 by matrix multiplication; is the $2^{O(|\varphi|)}$ translation necessary?

Answer Yes: G_k (size $O(k)$) needs $\geq 2^{k-1}$ compositions, even for circuits with complement and converse.

Difficulty Complement reverses inclusion; symmetric counter-models are infinite.

Method A finite colored graph with two witnesses per color pair; 2^k recolorings at one vertex; compare gate values by equality; each composition breaks the equality for ≤ 2 recolorings.

Further results in the paper: size-specific and randomized bounds; FO^3 vs. FO^2 sentences; matrix query languages over semirings.

Open: a superpolynomial gap over a fixed signature.

References i

E. F. Codd (1970). "A Relational Model of Data for Large Shared Data Banks". In: **Communications of the ACM** 13.6, pp. 377–387. DOI: 10.1145/362384.362685.

R. Brijder, F. Geerts, J. Van den Bussche, and T. Weerwag (2019). "On the Expressive Power of Query Languages for Matrices". In: **ACM Transactions on Database Systems** 44.4, 15:1–15:31.

A. Tarski and S. Givant (1987). **A Formalization of Set Theory without Variables**. Vol. 41. Colloquium Publications. Providence, RI: American Mathematical Society. DOI: 10.1090/coll/041.

Y. Nakamura and Y. Uezato (2026). "A Fine-Grained Dichotomy for Bounded-Variable Query Evaluation: The Calculus of Relations, a Boolean Modal Logic, and One-Variable Counting Logic". In: **Proceedings of the ACM on Management of Data** 4.5 (PODS). PODS 2027, 295:1–295:27. DOI: 10.1145/3850130.

Y. Nakamura (2022). "Expressive Power and Succinctness of the Positive Calculus of Binary Relations". In: **Journal of Logical and Algebraic Methods in Programming** 127, p. 100760. DOI: 10.1016/j.jlamp.2022.100760.

References ii

P. Jipsen, R. D. Maddux, and Z. Tuza (1995). "Small Representations of the Relation Algebra $E_{n+1}(1, 2, 3)$". In: **Algebra Universalis** 33, pp. 136–139. DOI: 10.1007/BF01190770.